

# The confinement scale in three bands, with published numbers, and the verification restated gauge-invariantly

Two corrections to the finite-verification statement of the Dobrushin note §4 and the finite-verification note, and a map of the region with numbers taken from the literature.

**Elitzur empties the single-link form of the block criterion.** Let  $V$  be a box of links and  $s$  a site all of whose links lie in  $V$ . The gauge transformation at  $s$  is a measure-preserving bijection of the Gibbs measure  $\mu_V^\omega$  for every boundary condition  $\omega$ , so the marginal of any link with an endpoint at  $s$  is invariant under all left (or right) translations of  $SU(3)$  and is therefore the Haar measure, whatever  $\omega$  is. The single-link influence  $\rho_V(x, y)$  vanishes for every link  $x$  with an interior endpoint, and the sum  $\sum_{x \in V} \sum_{y \notin V} \rho_V(x, y)$  receives contributions only from links with both endpoints on the boundary layer. That form of the criterion yields uniqueness and carries no information about the interior; the verification must use the **mixing form on sub-blocks**, which is the form equivalent to complete analyticity and the one that yields exponential decay: for every  $W \subseteq V$  and every  $y \notin V$ ,

$$\|\mu_V^\omega|_W - \mu_V^{\omega'}|_W\|_{\text{TV}} \leq C |W| e^{-\gamma d(W, y)} \quad (\omega, \omega' \text{ differing at } y).$$

Since the gauge-variant part of the marginal on  $W$  is uniform over the gauge orbit at the interior sites of  $W$ , the content of this condition is the conditional distribution of the gauge-invariant functions of the links in  $W$ , the traces of small Wilson loops, and that is how a verification should be organized. The single-link Dobrushin condition of the Dobrushin note §2 is unaffected, because it conditions on all other links and no gauge symmetry survives the conditioning.

**The boxes are small where the verification is needed.** With the scale  $r_0/a$  of Necco and Sommer (Nucl. Phys. B 622 (2002) 328) and the scalar glueball mass  $r_0 m_{0++} = 4.21$  of Morningstar and Peardon (Phys. Rev. D 60 (1999) 034509), both at metadata level and used for orientation only, the glueball correlation length in lattice units is

| $\beta_W$ | $g^2 = 6/\beta_W$ | $r_0/a$ | $m_{0++}a$ | $\xi/a$ | $\sigma a^2$ |
|-----------|-------------------|---------|------------|---------|--------------|
| 5.7       | 1.05              | 2.92    | 1.44       | 0.70    | 0.16         |
| 6.0       | 1.00              | 5.37    | 0.78       | 1.28    | 0.047        |
| 6.2       | 0.97              | 7.38    | 0.57       | 1.75    | 0.025        |
| 6.4       | 0.94              | 9.74    | 0.43       | 2.31    | 0.014        |

using  $r_0^2 \sigma \simeq 1.35$  for the string tension. At  $\beta_W = 5.7$  the glueball correlation length is below one lattice spacing, and a mixing verification would concern boxes of side 3 to 5, that is 216 to 2000 links, in place of the  $10^4$  to  $10^6$  links estimated earlier from a correlation length of 1 to 10. The slowest influence is carried by the flux-tube channel,  $\sigma a^2 \simeq 0.16$  per lattice unit at  $\beta_W = 5.7$ , which sets the box side rather than the glueball mass. Constants explicit where they are ours; nothing promoted.

*Caveat, 2026-09-23.* The  $m_{0++}a$  column scales the continuum ratio to each coupling. Near  $\beta_W \simeq 5.7$  the Wilson axis passes beside the endpoint of the fundamental–adjoint first-order line, where the scalar glueball softens (Heller 1995), so the lattice  $\xi/a$  there is likely larger than 0.70 and should be taken from the measured lattice mass (openings note, §1).

## 1. Three bands of the coupling

| band | $\beta_W$           | status                                                                                      |
|------|---------------------|---------------------------------------------------------------------------------------------|
| A    | $< 0.0135$          | gapped, proved by hand: Dobrushin single-link condition, $g^2 > 444$                        |
| B    | 0.0135 to about 5.7 | mixing with $\xi/a < 1$ by the published data; no rigorous method reaches it                |
| C    | above about 5.7     | $\xi/a$ grows, asymptotic scaling sets in near 6 to 6.5; the renormalization group's domain |

*Band B is a sharpness problem.* The theory there is, by the data, more strongly mixing than anything band A contains, yet no expansion converges with rigorous constants: the polymer expansion of the Wilson note reaches  $\beta_W \simeq 0.0057$ , the Dobrushin condition 0.0135, and an ideal entropy constant for plaquette surfaces, of order 10 to 15 in place of  $20e$ , would bring the expansion to  $\beta_W \sim 0.1$ . The strong-coupling series itself has singularities near  $\beta_W \simeq 3.5$  to 4, so even the true radius of convergence ends inside band B. The gap between 0.1 and 5.7 is a factor of 50 in  $\beta_W$  with no small parameter, in a regime where the correlation length is less than one lattice unit. This is the band the finite verification addresses, and Section 2 says what it costs.

*Band C is the renormalization group's.* From  $g^2 = 1/2$  ( $\beta_W = 12$ ) to  $\beta_W = 6$  ( $g^2 = 1$ ) is  $1/0.0966 \simeq 10$  one-loop doublings, over which  $\xi/a$  falls from its asymptotic-scaling value to about one. The stretch where  $\xi/a$  passes from about 10 to about 1 is three doublings wide at one loop. The physically nontrivial part of the problem, the place where the gap forms, is this stretch; everything in band B is, physically, already gapped with a gap of order  $\hbar c/a$ .

## 2. What the verification costs, restated

A mixing verification at  $\beta_W \simeq 5.7$  with boxes of side  $R = 3$  to 5 involves rigorous bounds on the conditional distribution of small Wilson loops inside  $V$ , with 216 to 2000 link variables, uniformly over the boundary condition, with the bound decaying in the distance from the boundary change at a rate consistent with  $\sigma a^2 \simeq 0.16$ . The uniformity over boundary conditions is a supremum over a compact set of dimension  $8|\partial V|$ , of order  $10^3$  to  $10^4$ , of a function defined by an integral over  $8|V|$  dimensions. No rigorous numerical method known to the author evaluates such a quantity with certified bounds today; the problem is nevertheless three to four orders of magnitude smaller in the number of variables than the earlier estimate, and it sits at a coupling where the theory's correlation length is below the lattice spacing, so that a certified bound on a single small box would settle band B at that coupling.

### 2b. The meeting point located: five order-one steps around $\beta_W \simeq 6$

The two unknown numbers of the finite-verification note §4 can be placed on the map. Read the three bands as blocking steps, one doubling of the lattice spacing each.

*Band C, from the weak side.* From  $g^2 = 1/2$  to  $\beta_W \simeq 6$  is ten one-loop doublings, and only the last three, in which  $\xi/a$  passes from about 10 to about 1, are outside the reach of a small-field expansion in principle; the first seven are the regime of the constructive programme, unavailable with constants but perturbative in kind.

*Band B, crossed by three steps.* The coarse plaquette at scale  $2^k a$  is a  $2^k \times 2^k$  Wilson loop of the fine lattice, whose expectation by the area law with  $\sigma a^2 \simeq 0.16$  at  $\beta_W = 5.7$  is about  $e^{-0.16 \cdot 4^k}$ : 0.53, 0.077,  $3.6 \times 10^{-5}$  for  $k = 1, 2, 3$ . Band A needs an activity below about  $2 \times 10^{-3}$ , so three doublings cross band B. The Migdal–Kadanoff recursion (Kadanoff, Ann. Phys. 100

(1976) 359; metadata level, a heuristic in four dimensions), which sends the activity to its fourth power per doubling, gives two and overestimates the decrease; the area-law count is the one to use. Rigorously, a block-spin step is well defined and its image is Gibbsian with a summable interaction in the strong-mixing regime (van Enter, Fernández and Sokal, J. Stat. Phys. 72 (1993) 879; metadata level), so once mixing is known at the top of band B the three steps that cross it are standard, and band A is proved by hand.

*The meeting point.* The verification is therefore needed at **one coupling**,  $\beta_W \simeq 5.7$  to 6, that is  $g^2 \simeq 1$ , where  $\xi/a \simeq 1$ , on boxes of side 3 to 5; its openness radius is the tolerance the renormalization group must meet, and everything below it follows. In the notation of the finite-verification note,  $g_{\text{DS}}^2 \simeq 1$  conditional on that single certified box, and the open requirement on the other side is  $g_{\text{RG}}^2 \geq 1$ : the small-field renormalization must be carried three doublings past the point where  $\xi/a \simeq 10$ .

**The whole problem in one sentence.** The mass gap for  $SU(3)$  is the control of about six consecutive blocking steps of the lattice theory at order-one coupling, uniformly in the volume: three from the weak side, where  $\xi/a$  falls from 10 to 1, and three from the strong side, which are standard once mixing is certified at the coupling in between. The conditional theorem states this as two hypotheses and an explicit conclusion. Everything else on the map is either perturbative in kind or proved with explicit constants.

### 3. What this changes

The finite verification is restated gauge-invariantly and at its true size. The decision it asks for is unchanged in kind and smaller in scale. Bands A and C are as before; band B is where a certified computation on a box of a few hundred links would connect the hand-proved region to the physical crossover, and band C remains the renormalization group's, ten doublings from  $g^2 = 1/2$  to the crossover at one loop. The glueball gap along the trajectory is then, in physical units,  $m_{0++} = 4.21/r_0 \simeq 1.66 \hbar c \text{ fm}^{-1}$  by the same data, which is what a proof would have to reproduce as a positive number.

### 4. Consequence for STATE

The Dobrushin note's specification is corrected: the block criterion in its single-link form is emptied by Elitzur and the mixing form on sub-blocks, organized around the conditional distribution of small Wilson loops, is the one to verify. The verification's size is corrected downward to boxes of side 3 to 5 at  $\beta_W \simeq 5.7$ , where  $\xi/a < 1$  by the published data. The three bands separate the sharpness problem (B, a factor 50 in  $\beta_W$  with  $\xi/a < 1$ ) from the physical problem (C, three doublings where  $\xi/a$  passes from 10 to 1).